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Quantifying Trust: Financial Risk Management for Trustworthy AI Agents

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PROBLEM

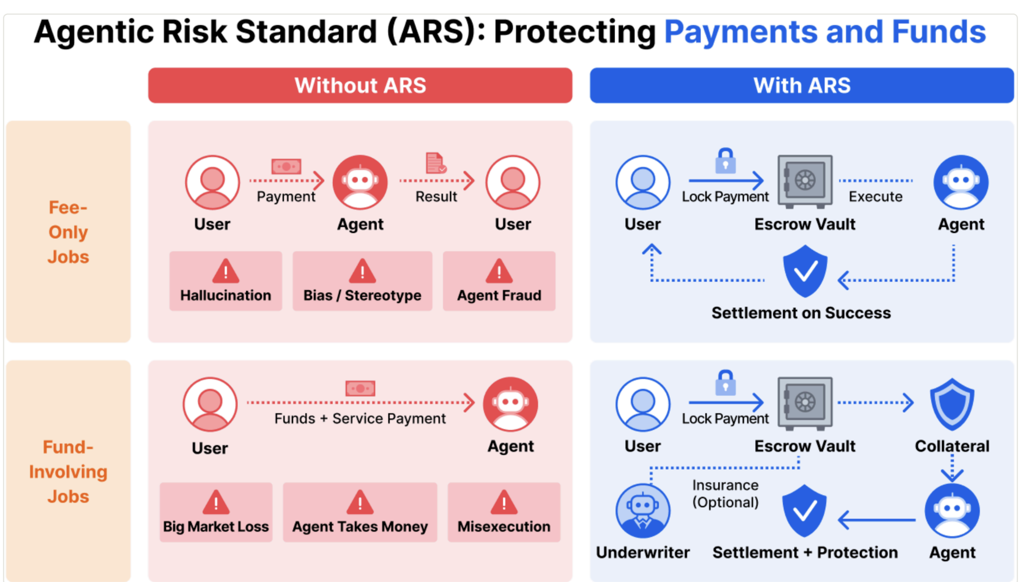
Agent behavior is inherently **stochastic**, so product-level harms — execution failure, misalignment, unintended outcomes — **cannot be eliminated** by model-internal safeguards like robustness or interpretability alone.

The operational meaning of trust shifts to **end-to-end outcomes** that technical safeguards cannot guarantee.

MOTIVATION

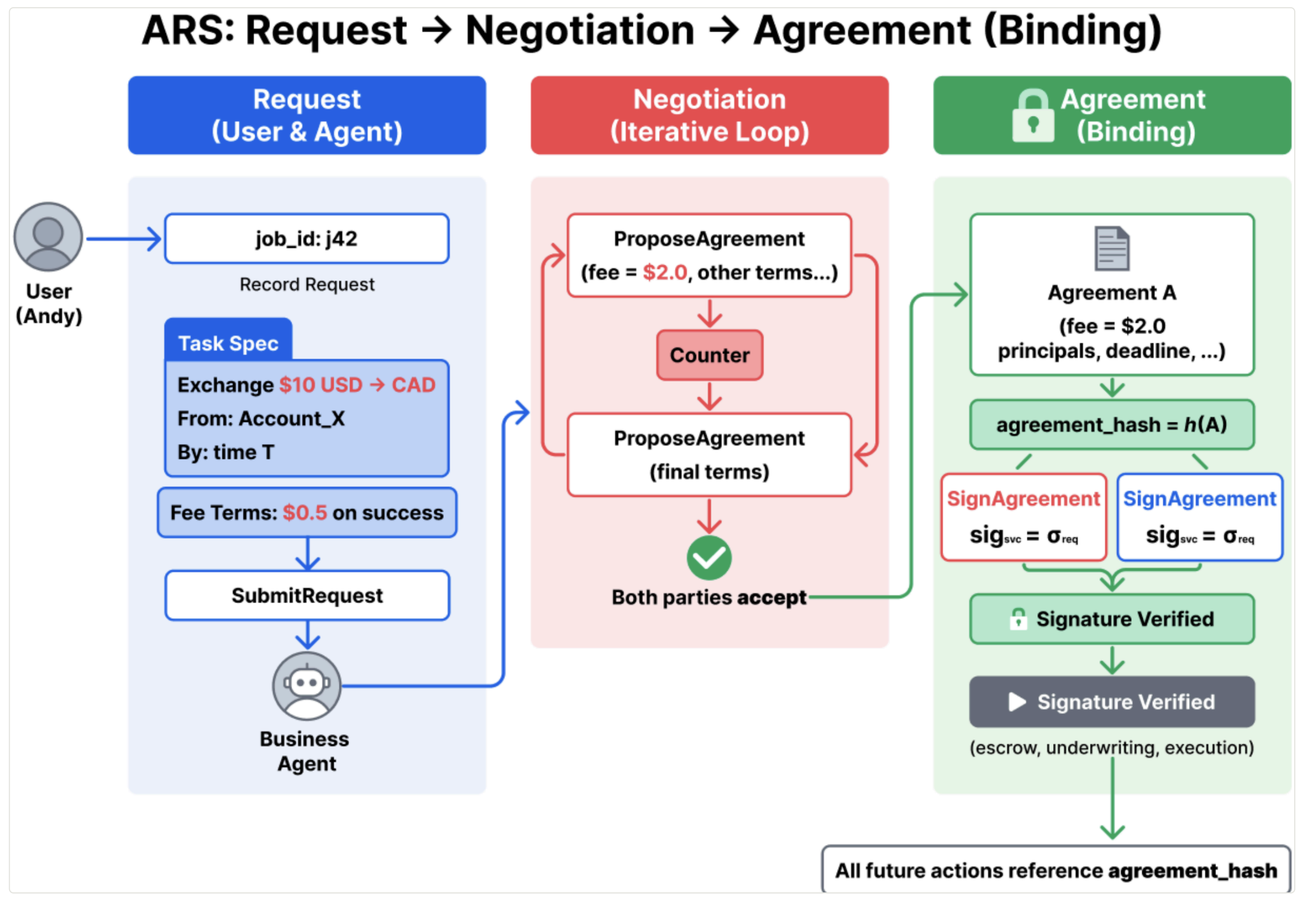
> *High-stakes industries never rely on trust alone — they use insurance and underwriting to price and absorb risk.*

- Prior trustworthy-AI work treats safety as a *model property*; none give users **predefined, contractually enforceable recourse** when a transaction fails.
- ARS borrows **financial underwriting** to make agentic-transaction risk bearable and priced.



Without ARS, users prepay agents and absorb every downstream harm; with ARS, fees lock in escrow and release only on verified success, with optional underwriting for fund-moving jobs.

METHOD



ARS job lifecycle: Request → Negotiation → binding Agreement anchored by a signed agreement_hash.

1 Standard-Mediated Job Lifecycle

transaction layer

- A **transaction-layer settlement standard** uniting risk assessment, underwriting, escrow, and compensation.
- A **requestor**, **business agent**, and **underwriter** interact through a typed action vocabulary, each one bound to a signed **agreement_hash** for audit.

3 standard roles, one auditable transaction interface

2 End-to-End Transaction Flow

5 stages

Request → **Assess** → **Collateral** → **Execute** → **Settle**

ARS targets **pre-verification fund exposure**, where principal releases before verification, so fee escrow alone cannot protect the buyer.

> *Every action stays attributable to a standard role and bound to the signed agreement.*

3 Underwriting: Collateral & Premium

actuarial pricing

For principal M, a noisy estimate $\hat{p}_{uw} = p(1-f_n) + (1-p)f_p$ drives a **sigmoid collateral** $D = \sigma(\hat{p}_{uw})\,M$ and an actuarially-fair premium with loading λ:

$$\Pi = \hat{p}_{uw}\,(1 - \sigma(\hat{p}_{uw}))\,M\,(1+\lambda)$$

residual exposure after collateral, scaled by the loading factor

Midpoint m sets where collateral reaches $0.5M$; steepness s controls how sharply low- and high-risk jobs separate.

4 Three-Stage Friction Gate

protection vs. participation

- **Sigmoid sizing** sharply separates low- vs. high-risk jobs while staying continuous — near-zero collateral for safe jobs, approaching full principal as risk rises.
- **Merchant posting**: willingness to post collateral falls linearly from 0.90 at zero collateral to 0.10 at full collateral, capturing real participation friction.
- **Human override** proceeds without coverage (probability 0.50) — no premium is collected and the buyer bears the full loss on failure, so the gate balances protection against participation.

Two complementary protections: **contingent reimbursement** for covered failures plus an **ex-ante collateral buffer** that shifts downside to the business agent.

The premium is **actuarially consistent** with collateral: as collateral nears full principal, residual exposure and premium both vanish, while risky uncollateralized jobs pay more. On resolution, collateral is **returned** when no failure occurs; on failure it is forfeited up to realized loss and the underwriter reimburses the rest.

KEY RESULTS

LOADING Λ	LOSS REDUCTION	ADOPTION
0.0	**61%**	**0.446**
0.5	~42%	0.30
1.0	24%	0.126

SO WHAT → *Loss reduction stays **positive across the whole sweep**; collateral deters risky jobs before execution.*

HEADLINE NUMBERS

61% Peak reduction in mean user loss
vs. no underwriting (zero loading)

15-20% failure events deterred

5,000 simulation episodes

$9.4K peak underwriter wallet

TAKEAWAY

Treating agent reliability as an **underwritable financial risk** — not a model property — gives enforceable, priced protection

SO WHAT

Trust becomes a priced, enforceable guarantee.

SCAN TO READ

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github.com/t54-labs/AgenticRiskStandard

Paper

Code